Updated DGP Simulation

August 14, 2026

1 Question, notation, and estimands

The proposal's wage interface is

$$Y_{ij} = X_{ij}'\delta + \alpha_i + \psi_j + u_i'\Lambda v_j + \varepsilon_{ij}, \qquad \text{rank}(\Lambda) = r. \tag{1}$$

Simulation implementation:

- Set $X_{ij}'\delta = 0$ to isolate worker, firm, and interaction terms.
- Define the systematic schedule as $m_{ij} = \mathbb{E}[Y_{ij} \mid i, j]$.
- Let J_{it} be worker i's employer in period t and observe $Y_{it} = m_{iJ_{it}} + \varepsilon_{it}$.
- Apply every procedure under every DGP. An additive-only design would favor additive methods by construction.
- In each replication, draw a population and panel, run all procedures, and score them against that population's exact truth.

Table 1 defines every symbol used below. A prime denotes transposition. All expectations and variances are finite sums, not large-population approximations.

Table 1: Notation

Symbol	Definition	Interpretation
i, j, t	Worker, firm, and period indices	Indices used in the proposal and simulated panel
Y_{ij}, Y_{it}, J_{it}	Proposal wage, observed panel wage, and employer	$Y_{it} = m_{iJ_{it}} + \varepsilon_{it}$ in the simulation
m_{ij}	$X_{ij}'\delta + \alpha_i + \psi_j + u_i'\Lambda v_j$	Complete systematic worker–firm wage schedule
$\alpha_i, \psi_j, u_i, v_j, \Lambda$	Proposal components in Equation (1)	Worker effect, firm effect, factors, and interaction matrix
P_{ij}^{obs}	Population probability of match (i, j)	Observed assignment law, with $\sum_{ij} P_{ij}^{\text{obs}} = 1$
p_i, q_j	$\sum_j P_{ij}^{\text{obs}}$ and $\sum_i P_{ij}^{\text{obs}}$	Worker and firm population weights
$p_i q_j$	Product of the two marginals	Counterfactual independent assignment
$\mathbb{E}_{P^{\text{obs}}}, \mathbb{E}_{pq}$	Expectations under P_{ij}^{obs} and $p_i q_j$	Observed and independent-assignment averages
a_i, b_j, h_{ij}	$m_{ij} = \mu + a_i + b_j + h_{ij}$ under $p_i q_j$	Worker main effect, firm main effect, and interaction
r	Number of nonzero factor directions in h_{ij}	Interaction rank; $r = 0$ is additive
Q_F	$\mathbb{E}_{pq}[(m_{ij} - \mathbb{E}_q[m_{ij} \mid i])^2]$	Firm-side schedule variation
H_F	$2\mathbb{E}_{pq}[h_{ij}^2]$	Twice the interaction variance
ρ_H	$H_F/(2Q_F)$ when $Q_F > 0$	Interaction share of firm-side variation
C_{assign}^w	$\{\text{Var}_{P^{\text{obs}}}(m) - \text{Var}_{pq}(m)\}/2$	Proposal assignment contribution to wage-schedule variance
$\alpha_i^{\text{AKM}}, \psi_j^{\text{AKM}}$	Pseudo-true additive worker and firm effects	Native AKM/KSS projection objects
$\lambda_i^{\text{BS}}, \mu_j^{\text{BS}}$	$\mathbb{E}_{P^{\text{obs}}}[m_{ij} \mid i]$ and $\mathbb{E}_{P^{\text{obs}}}[m_{ij} \mid j]$	Native BS20 worker and firm wage types
$\rho_{\text{BS}}, C_{\text{BS}}$	Correlation and covariance of the two BS20 wage types	Native BS20 sorting summaries

For any completed schedule, use the product-weighted ANOVA decomposition

$$\mu = \mathbb{E}_{pq}[m_{ij}], \qquad a_i = \mathbb{E}_q[m_{ij} \mid i] - \mu, \tag{2}$$

$$b_j = \mathbb{E}_p[m_{ij} \mid j] - \mu, \qquad h_{ij} = m_{ij} - \mu - a_i - b_j. \tag{3}$$

The weighted margins of h_{ij} are zero. Therefore

$$Q_F = \text{Var}_q(b_j) + \mathbb{E}_{pq}[h_{ij}^2] = \text{Var}_q(b_j) + \frac{H_F}{2}. \tag{4}$$

Equation (4) follows by substitution into the definition of Q_F and the zero weighted margins of h_{ij}. In a centered low-rank DGP, (a_i, b_j, h_{ij}) correspond to $(\alpha_i, \psi_j, u_i'\Lambda v_j)$; they do not replace the proposal symbols.

Interpretation. Q_F measures all firm-side schedule variation; H_F isolates the part that changes with the worker; ρ_H is its unit-free share; C_{assign}^w is the variance change from observed rather than independent matching, holding worker and firm marginals fixed.

Reproducibility. The merged output contains 33,900 scalar estimates and 4,300 procedure attempts. Configuration fingerprint:

```
e075e50934629f7ae20971667f117c418f3628bf576653f079602a07eb76a735
```

All reported numbers come from that output. External claims are cited; uncited equations are definitions, derived identities, or declared design choices.

2 Simulation design

2.1 Common population, assignment, and panel

Fixed settings:

- **Size:** $N = 25{,}000$ workers, $J = 5{,}000$ firms, $T = 60$ periods; 1.5 million observations before procedure-specific cleaning.
- **Why 25,000 workers:** the lower end of the proposed 25,000–50,000 range keeps cluster cost feasible and gives five workers per firm on average. The 5,000-firm count was fixed in advance.
- **Weights:** uniform worker and firm population marginals.
- **Noise:** independent $\varepsilon_{it} \sim N(0, 0.5^2)$. The standard deviation is one half of the standardized main-effect standard deviation.
- **Seeds:** master seed 20260812; separately derived population, panel, and estimator seeds by replication.
- **Normalization:** Gaussian main effects have finite-population mean zero and variance one. Factor columns are centered and weighted orthonormal. Reported singular values therefore hold in each population.

Observed assignment probabilities are

$$P_{ij}^{\text{obs}} = R_i S_j \exp\left\{ 0.5 a_i b_j + 0.3 \frac{h_{ij}}{\sqrt{\mathbb{E}_{pq}[h_{ij}^2]}} \right\}, \tag{5}$$

where the interaction term is omitted when $h_{ij} = 0$. Interpretation:

- 0.5 controls sorting on common worker and firm components.
- 0.3 controls sorting on match-specific gains.
- Positive R_i, S_j restore uniform marginals. Thus C^w_{assign} changes only with the joint matching pattern, not with worker or firm composition.
- The coefficients create visible sorting without nearly deterministic matching.

Mobility. Let $\pi_{ij} = P_{ij}^{\text{obs}}/p_i$. Draw the initial firm from π_i. Later, stay with probability 0.6 or redraw from π_i with probability 0.4. A redraw may return the same firm, so realized retention is slightly above 0.6. This gives about as many redraw opportunities as 120 periods with 0.8 retention, using a shorter and more mobile panel.

Design lock and cluster run. A one-replication mobility gate verified full-population retention and a numerical pass for rank two. The 60-period design was then fixed before the 100-replication accuracy results were viewed. The run used 50 independent shards; the merge checked the fingerprint, replication indices, and duplicates. One hundred replications balance cost with detection of the rare failures missed by a small pilot.

Exact truth. Each complete $25{,}000 \times 5{,}000$ schedule and assignment matrix is known. Project truths use all 125 million worker–firm cells before panel sampling; they are not estimates from observed matches.

2.2 Five wage schedules

Table 2: DGP-specific parameters

DGP	Systematic wage schedule	Declared interaction
AKM	$m_{ij} = \alpha_i + \psi_j$ with independent raw Gaussian effects	Rank 0 [1]
Crippa/Tukey	$m_{ij} = \alpha_i + \psi_j + 0.75\alpha_i\psi_j$	Rank 1; singular value 0.75 [4, Sec. 2.2]
BLM types	Two approximately equal worker types and three approximately equal firm classes; $m_{ij} = A_{G_i} + B_{K_j} + u_{G_i}v_{K_j}$ after normalization	Rank 1; singular value 1 [2]
Low-rank factors	$m_{ij} = \alpha_i + \psi_j + u_i' \operatorname{diag}(1, 0.5)v_j$; raw worker-factor correlation with α_i is 0.35 and raw firm-factor correlation with ψ_j is 0.25	Rank 2; singular values $(1, 0.5)$
GKLP	$m_{ij} = Z_i + c_j + b_j\eta_i + \frac{1}{2}b_j^2(0.5)^2$, recentered; raw correlations are $\operatorname{Corr}(Z_i, \eta_i) = 0.35$ and $\operatorname{Corr}(c_j, b_j) = 0.25$	Rank 1; comparative-advantage interaction $b_j\eta_i$ [5, 6]

Underlying draws:

- **AKM and Crippa/Tukey:** independent raw worker and firm effects from $N(0, 1)$.
- **Continuous low rank:** raw factors $\widetilde{u}_{i\ell} = 0.35\widetilde{\alpha}_i + \sqrt{1 - 0.35^2}\, z_{i\ell}$ and $\widetilde{v}_{j\ell} = 0.25\widetilde{\psi}_j + \sqrt{1 - 0.25^2}\, z_{j\ell}$; innovations are standard normal before normalization.
- **GKLP:** (Z_i, η_i) and (c_j, b_j) are jointly standard Gaussian with correlations 0.35 and 0.25.
- **BLM:** worker and firm labels are as evenly sized as possible, then shuffled. Type main effects and raw type factors are independent standard Gaussian draws before weighted normalization. The cell table is redrawn in every replication.

Purpose of each DGP:

- **AKM:** additive implementation check.
- **Crippa/Tukey:** simplest smooth rank-one nonadditivity.
- **BLM:** exact specification for the grouped procedure.
- **Low-rank factors:** direct rank-two implementation of the proposal interface.
- **GKLP:** economic comparative advantage; low rank but not built from the project estimator.

The Crippa and GKLP functional forms are adapted from the cited papers. Normalization, sorting coefficients, and distributions are simulation choices. DGP code is separate from estimator code: no estimator receives latent truth or the complete schedule.

Table 3 gives target scale before estimation error. Normalization nearly fixes Q_F, H_F, and ρ_H within a DGP; C^w_{assign} varies with the realized sorting kernel.

Table 3: Mean exact finite-population proposal truths across 100 replications

DGP	Q_F	H_F	ρ_H	C^w_{assign}
AKM	1.0000	0.0000	0.0000	0.4142
Crippa/Tukey	1.5625	1.1250	0.3600	0.6390
BLM types	2.0000	2.0000	0.5000	0.3728
Low-rank factors	2.2500	2.5000	0.5556	0.4724
GKLP	2.0313	2.0000	0.4923	0.5923

3 Procedures and evaluation rules

3.1 KSS-HE

KSS corrects plug-in variance components in linear models with many fixed effects; KSS-HE allows unrestricted heteroskedasticity [7]. Its native target is the assignment-weighted additive projection

$$(\mu, \alpha, \psi) = \arg\min_{\mu,\alpha,\psi} \mathbb{E}_{P^{\text{obs}}}[(m_{ij} - \mu - \alpha_i - \psi_j)^2]. \tag{6}$$

Implementation and mapping:

- **Software:** PyTwoWay 0.3.21 with approximate trace and leverage calculations (`fe_exact=False`) [8]. Exact leverage is impractical at 1.5 million observations; the setting is fixed across DGPs.
- **Native outputs:** $V_\psi^{\text{AKM}} = \text{Var}_q(\psi_j^{\text{AKM}})$ and $C_{\psi\alpha}^{\text{AKM}} = \text{Cov}_{P^{\text{obs}}}(\alpha_i^{\text{AKM}}, \psi_j^{\text{AKM}})$.
- **Nonadditivity:** these native objects need not equal Q_F and C^w_{assign}. KSS imposes $H_F = \rho_H = 0$; these are structural zeros, not estimated interaction moments.
- **Displayed version:** KSS-HE, because it makes the weaker error-variance assumption. AKM-FE and KSS-HO remain in the CSV files.
- **Cleaning:** leave-one-spell connected cleaning retains all 1.5 million observations.

3.2 BLM

BLM uses discrete worker types and firm classes [2]. This implementation uses PyTwoWay 0.3.21 [8]:

- **Model size:** two worker types and three firm classes. This is correct under the BLM DGP and deliberately coarse under continuous DGPs.
- **Firm clustering:** wage-CDF features use all 60 periods.
- **Likelihood sample:** periods 0 and 1 only; stayer if $j_{i0} = j_{i1}$, mover otherwise. Later moves do not alter this label. This avoids the earlier error of restricting stayers to workers who never move in 60 periods.
- **Optimization:** ten CDF points, four initializations, best two retained starts, 250-iteration cap, and tolerance 10^{-6}; fixed across DGPs.
- **Support:** require all three stayer classes and nine mover origin–destination pairs. All 500 samples pass. Each fit uses 50,000 static-period observations.
- **Scoring:** after estimation, permute the 2×3 table to minimize cell-mean squared error against a known evaluation table. Under BLM, use simulated groups; under continuous DGPs, use deterministic equal-count groups ordered by product-marginal mean wage. Outside the BLM DGP, error against the full truth therefore includes six-cell approximation error.

3.3 BS20

BS20 estimates moments of worker and firm wage types [3]. Its native covariance is

$$C_{\mathrm{BS}} = \mathrm{Cov}_{P^{\mathrm{obs}}}(\lambda_i^{\mathrm{BS}}, \mu_j^{\mathrm{BS}}), \qquad \rho_{\mathrm{BS}} = \mathrm{Corr}_{P^{\mathrm{obs}}}(\lambda_i^{\mathrm{BS}}, \mu_j^{\mathrm{BS}}), \tag{7}$$

where $\lambda_i^{\mathrm{BS}} = \mathbb{E}_{P^{\mathrm{obs}}}[m_{ij} \mid i]$ and $\mu_j^{\mathrm{BS}} = \mathbb{E}_{P^{\mathrm{obs}}}[m_{ij} \mid j]$.

- Apply PyTwoWay's weighted estimator after strongly connected, spell-level, no-return cleaning [8]; about 1.491 million observations remain on average.
- Read `cov(lambda, mu)` directly. It is not constructed from project quantities.
- Since $C_{\mathrm{BS}} \neq C^w_{\mathrm{assign}}$ in general, score BS20 against ρ_{BS} and C_{BS} and show the estimand gap. Do not report it as an estimator of C^w_{assign}.

3.4 Project low-rank plug-in

The current project benchmark fits

$$m_{ij} = \alpha_i + \psi_j + u_i' \Lambda v_j, \qquad \mathrm{rank}(\Lambda) = r. \tag{8}$$

Implementation:

- Collapse periods to worker–firm match means; weight by match counts.
- Fit positive ranks by alternating least squares (ALS): six exploratory starts, then three perturbed confirmation starts around the best basin.
- Center and SVD-normalize after every ALS iteration. Thus the numerical tail below is not caused by omitted centering.
- Use tolerance 10^{-6} and a 300-iteration cap.
- At rank r, require $r + 2$ distinct firm partners: one above the algebraic $r + 1$ minimum. Use degree-three support for candidates $\{0, 1\}$ and degree four for $\{0, 1, 2\}$.
- Retain all 25,000 workers, 5,000 firms, and 1.5 million observations.

BIC selects the rank:

$$\mathrm{BIC}(r) = n \log(\mathrm{SSE}_r / n) + \mathrm{df}(r) \log n, \qquad \mathrm{df}(r) = N + J - 1 + r(N + J - 2 - r). \tag{9}$$

- Candidates are $\{0, 1\}$ under AKM and BLM and $\{0, 1, 2\}$ under Crippa, continuous low rank, and GKLP.
- Each set contains the DGP rank and an underfitted alternative. Rank two is impossible with two BLM worker types; ranks above two are outside scope.
- Equation (9) is a simulation choice, not a result from the unfinished LOO theory.
- The selected fit completes unobserved cells, then applies the project definitions. It is a plug-in without the leave-worker-out correction.

3.5 Completed calibration versus untested estimator variants

Table 4 separates baseline choices from untested estimator variants. A successful preflight does not show that a choice lowers unconditional bias or RMSE.

Table 4: What was tried, and what remains untested

Choice	Production status	What the evidence does and does not establish
Center and SVD-normalize factors after every ALS iteration	Used	Fixes the factor normalization and prevents arbitrary location or rotation drift. It was not a separate comparison arm and did not eliminate warnings.
Require at least $r+2$ distinct partners on the common support core	Used	Adds one partner beyond the algebraic $r+1$ minimum. It is a declared support safeguard, not ridge regularization.
Six exploratory starts plus three confirmation starts	Used	Searches more basins and checks whether the best basin reproduces. This makes instability visible; it does not force unstable local regressions to become stable.
Sixty periods with redraw probability 0.40	Used	Increases mobility relative to the failed high-retention preflights. This changes the simulated information set, not the estimator.
Ridge or Tikhonov local factor regressions	Not run	Proposed estimator variant. No claim about its bias, RMSE, or warning rate is supported by the current results.
Condition-number-adaptive penalty, pooling, or support rule	Not run	Proposed estimator variant. The rule and tuning parameter must be declared before examining the returned functionals.
Direct estimation of Q_F, H_F, and C^{w}_{assign} without full completion	Not run	Proposed way to avoid extrapolating every unobserved worker–firm pair.
Full leave-worker-out correction	Not run	Still under development and intentionally excluded from the production comparison.

- **"Conditioning as part of the estimator":** before each update, inspect the local Gram matrix. If independent local variation is weak, apply a prespecified ridge, pooling, or support rule. The rule changes returned estimates by design.
- **Not the same as pass-only:** pass-only drops warning fits after estimation; it does not modify the estimator.
- **Required next experiment:** fix DGP draws and seeds; compare the baseline with declared variants using unconditional bias, RMSE, warning rate, and any support-induced change in the target population.
- **Current evidence:** the variants are research directions, not simulation findings.

3.6 Mappings, status, and accuracy metrics

Table 5 prevents unlike objects from being relabeled. "Cross-target" means deliberate comparison of a native output with a different project quantity. A dash means no defensible mapping.

Table 5: Mapping from procedure outputs to project quantities

Procedure	Q_F	H_F	ρ_H	C^w_{assign}
KSS-HE	Native firm variance; cross-target	0, imposed	0, imposed	Native additive covariance; cross-target
BLM	Grouped schedule	Grouped schedule	Grouped schedule	Grouped schedule
BS20	–	–	–	–; native outputs are ρ_{BS} and C_{BS}
Project plug-in	Completed schedule	Completed schedule	Completed schedule	Completed schedule

For replication s, define $e_s = \widehat{\theta}_s - \theta_s$. Among S returned values,

$$\text{Bias} = S^{-1} \sum_s e_s, \qquad \text{RMSE} = \sqrt{S^{-1} \sum_s e_s^2}, \qquad \text{MCSE(Bias)} = \frac{\text{sd}(e_s)}{\sqrt{S}}. \tag{10}$$

Metric interpretation:

- **Bias:** direction of average error.
- **RMSE:** main accuracy measure; combines bias and dispersion and penalizes large errors. Compare procedures within an estimand because scales differ.
- **MCSE(bias):** Monte Carlo noise in reported bias, not empirical standard error in one data set.

Status rules:

- **KSS and BS20 pass:** PyTwoWay returns a value.
- **BLM pass:** complete static support and no one-step likelihood decline above 10^{-4}.
- **Project pass:** best fit converges. At positive rank, at least two starts lie within $0.0001 \max(1, O^*)$ of the best objective O^*, and near-best spreads in Q_F, H_F, and C^w_{assign} are at most $0.001 \max(1, |\theta^*|)$.
- The project thresholds check numerical reproducibility; they are not statistical critical values. ρ_H is derived from Q_F and H_F, so it does not receive a separate spread test.
- **Headline:** all-returned RMSE. Pass-only RMSE is a diagnostic sensitivity calculation, not a replacement estimator.

Display order and purpose:

1. **Status:** numerical reliability.
2. **Own-model benchmarks:** implementation under correct specification.
3. **Common-target RMSE:** comparison of defensibly mapped procedures.
4. **Native-minus-project gaps:** estimation error versus target difference.
5. **Rank selection:** model-dimension performance.
6. **All-returned versus pass-only RMSE:** warning-related tail risk.

4 Results

4.1 Completion and numerical status

Table 6 and Figure 1 report, in order: pass or completion, warning, unsupported, and failure.

Table 6: Status counts: pass or completion / warning / unsupported / failure

DGP	KSS	BLM	BS20	Project
AKM	100/0/0/0	100/0/0/0	100/0/0/0	100/0/0/0
Crippa/Tukey	100/0/0/0	100/0/0/0	100/0/0/0	39/61/0/0
BLM types	100/0/0/0	100/0/0/0	100/0/0/0	85/15/0/0
Low-rank factors	100/0/0/0	100/0/0/0	100/0/0/0	93/7/0/0
GKLP	100/0/0/0	100/0/0/0	100/0/0/0	100/0/0/0

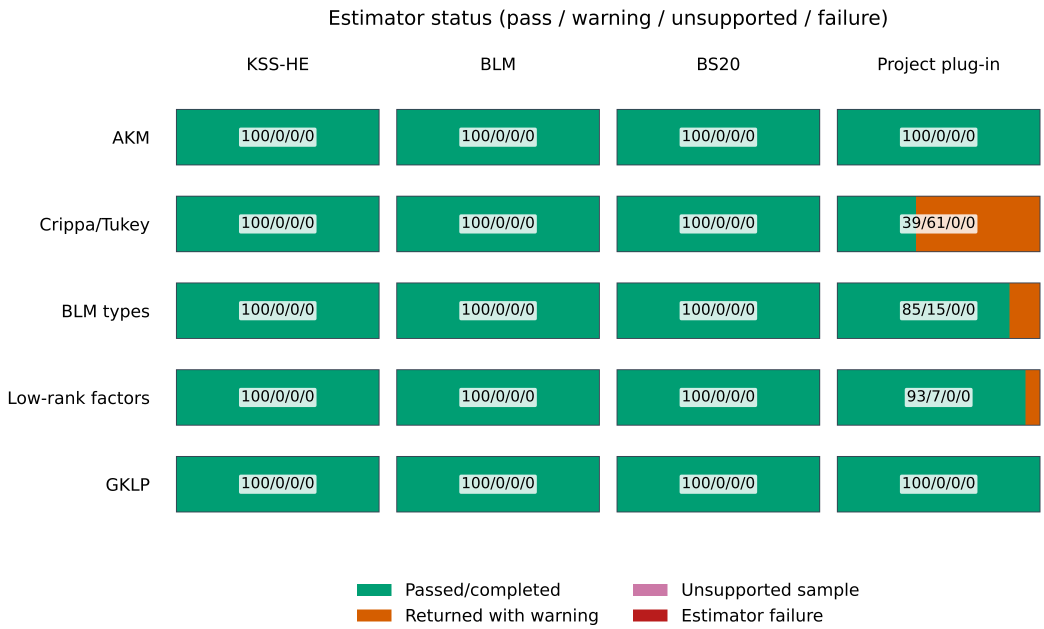

Figure 1: The status mosaic precedes accuracy because error magnitudes cannot reveal numerical unreliability. KSS and BS20 use completion only; BLM and the project plug-in also use the Section 3 diagnostics.

Key facts:

- KSS, BLM, and BS20 complete all 500 attempts.
- The corrected two-period BLM sample has no unsupported samples and no optimizer failures; the earlier permanent-stayer error is removed.
- The project plug-in passes all AKM and GKLP replications. Warnings: 61 Crippa, 15 BLM, and 7 continuous low rank.
- Warning estimates are finite and remain in unconditional RMSE.
- A warning need not imply a large economic error. Under Crippa, 54 of 61 warnings fail only the Q_F spread test, often near its 0.001 boundary. Section 4.6 checks whether warnings identify tail risk.

4.2 Correctly specified benchmarks

Correctly specified checks:

- **KSS under AKM:** Q_F bias 0.00008 (MCSE 0.00025), RMSE 0.00250; C^w_{assign} bias -0.00011

(MCSE 0.00015), RMSE 0.00145.

- **BLM under BLM:** truth $(Q_F, H_F, \rho_H, C^w_{\text{assign}}) = (2, 2, 0.5, 0.37285)$. Bias (MCSE) is 0.00574 (0.00952), -0.01443 (0.00708), -0.00403 (0.00248), and -0.00126 (0.00217), respectively.
- These results validate the KSS and corrected BLM implementations before interpreting misspecification elsewhere.

Table 7 also reports the project plug-in. All-returned rows measure procedure performance. Pass-only rows condition on the numerical diagnostic.

Table 7: RMSE against the full project truth in key benchmark rows

Procedure	DGP	Pass	Q_F	H_F	ρ_H	C^w_{assign}
KSS-HE	AKM	100	0.0025	0*	0*	0.0015
BLM	BLM types	100	0.0949	0.0719	0.0250	0.0216
Project	AKM	100	0.0027	0*	0*	0.0015
Project	Crippa/Tukey	39	94.2143	45.9700	0.0515	47.1163
Project, pass only	Crippa/Tukey	39	0.0116	0.0198	0.0038	0.0060
Project	BLM types	85	63.1342	126.1343	0.0897	43.7128
Project, pass only	BLM types	85	0.0219	0.0425	0.0050	0.0139
Project	Low-rank factors	93	26.4746	48.9358	0.0938	13.2346
Project, pass only	Low-rank factors	93	0.0171	0.0304	0.0027	0.0053
Project	GKLP	100	0.0092	0.0156	0.0018	0.0040

- Project plug-in all-returned RMSE is small under AKM and GKLP.
- Pass-only RMSE is also small under Crippa, BLM, and continuous low rank.
- This locates the problem in a numerical tail; it does not repair unconditional performance.

4.3 Accuracy for common project targets

Figure 2 applies each defensible mapping in Table 5 to the same project truth. The color scale is logarithmic because values range from below 10^{-28} (AKM structural zeros) to above 100 (project warning fits). Printed values are not clipped. BS20 is N/A because C_{BS} is not a project estimand.

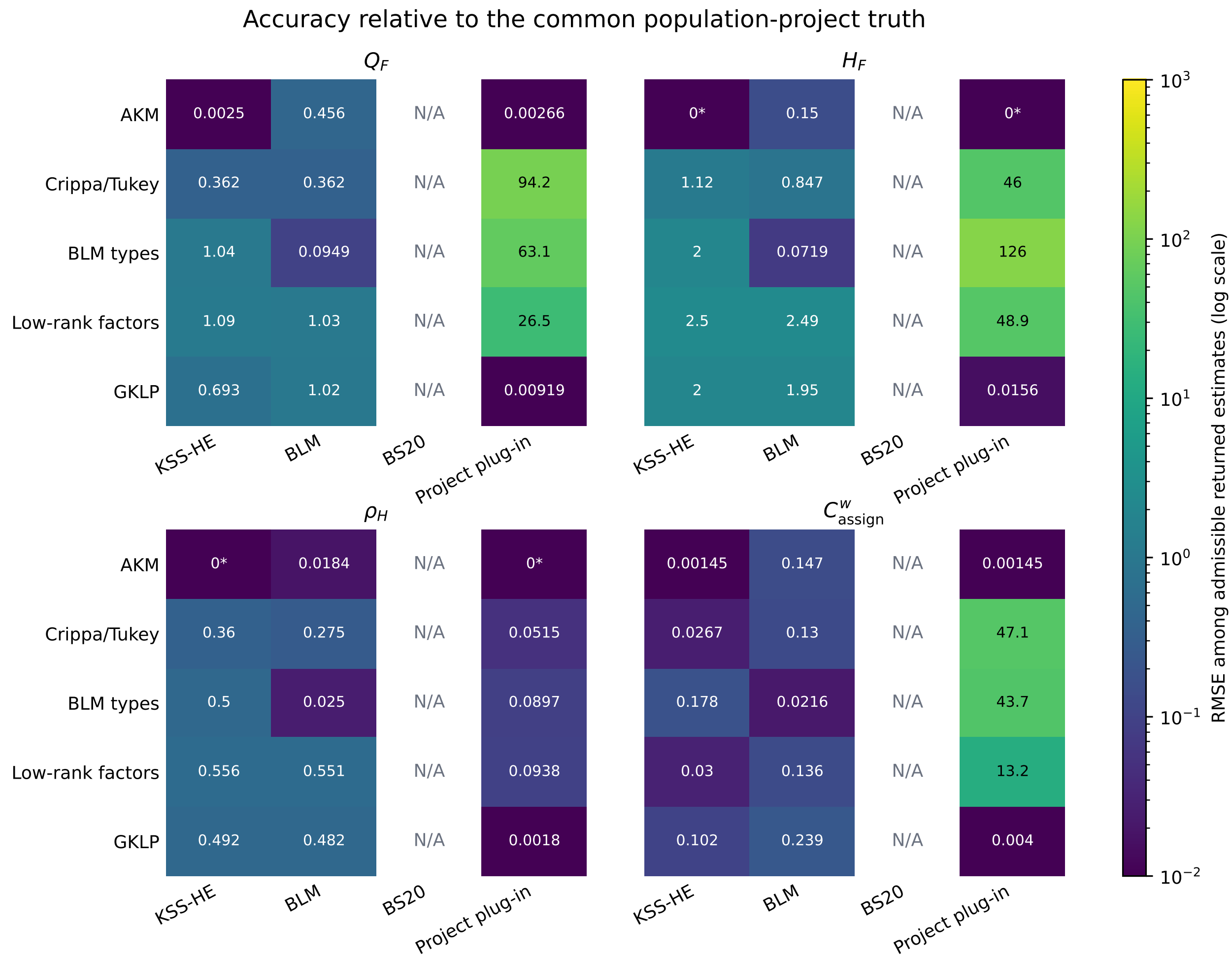

Figure 2: RMSE among all admissible returned estimates against the common project truth. The heatmap is the main symmetric comparison: every DGP is a row, every mapped procedure a column, and each panel holds one estimand. N/A means no defensible mapping; 0* denotes an AKM structural zero.

Read the heatmap in three steps:

- **KSS:** accurate for additive native objects. Under nonadditivity, imposed $H_F = \rho_H = 0$ makes common-target RMSE equal omitted truth: $(1.125, 0.36)$ under Crippa and $(2.5, 0.556)$ under continuous low rank.
- **BLM:** accurate under its discrete DGP. Six cells lose most continuous interaction variation; continuous-low-rank H_F RMSE is 2.490.
- **Project plug-in:** GKLP RMSEs are $(0.00919, 0.01565, 0.00180, 0.00400)$. Rare completion failures dominate unconditional RMSE under the other three nonadditive DGPs.

4.4 Native targets versus project targets

For procedure p, the common-target error decomposes exactly as

$$\widehat{\theta}_p - \theta_{\text{proj}} = (\widehat{\theta}_p - \theta_p) + (\theta_p - \theta_{\text{proj}}). \tag{11}$$

The two terms are:

- $\widehat{\theta}_p - \theta_p$: native estimation error;

- $\theta_p - \theta_{\text{proj}}$: estimand gap, present before panel sampling.

Figure 3 plots the second term. Subtraction isolates a difference in definitions; it does not subtract an estimate from an arbitrary quantity.

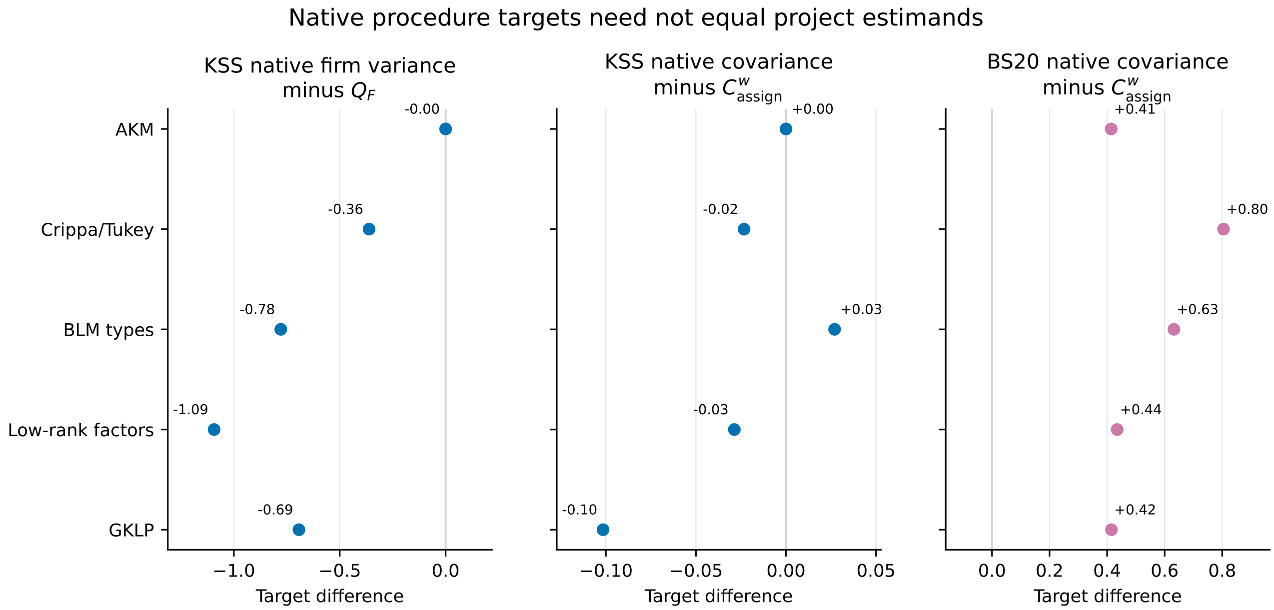

Figure 3: Population estimand differences averaged across the 100 populations in each DGP. The display is needed because common-target RMSE otherwise mixes estimation error with disagreement about the target itself.

KSS.

- Native Q_F gaps are about: AKM, 0; Crippa, -0.361; BLM, -0.778; low rank, -1.093; GKLP, -0.693.
- Native Q_F bias is only -0.00053 to 0.00061. Thus the main common-target error under nonadditivity is omitted interaction variation, not poor estimation of the additive projection.
- Under independent assignment, Equation (4) gives $\text{Var}_q(b_j) - Q_F = -H_F/2 \leq 0$. With sorted assignment, no universal sign result holds; the negative simulated signs follow from this design.

BS20.

- Native covariance exceeds C^w_{assign} by 0.414 to 0.805; native covariance RMSE is only 0.00867 to 0.02960.
- In the additive case, define $Tb(i) = \mathbb{E}[b_J \mid I = i]$ and $T^*a(j) = \mathbb{E}[a_I \mid J = j]$. Expanding Equation (7) gives

$$C_{\text{BS}} - C^w_{\text{assign}} = \|Tb\|_p^2 + \|T^*a\|_q^2 + \langle Tb, TT^*a\rangle_p \geq 0. \tag{12}$$

The inequality uses contraction of conditional expectation and $x^2 + y^2 - xy \geq 0$. It explains the positive AKM gap. No corresponding universal sign result holds for a general nonadditive schedule.

Table 8 is separate from the common-target heatmap because it tests BS20 against ρ_{BS} and C_{BS} without implying that either equals C^w_{assign}.

Table 8: BS20 accuracy against its native correlation and covariance targets

DGP	ρ_{BS} truth	Bias	RMSE	C_{BS} truth	Bias	RMSE
AKM	0.4142	-0.0024	0.0028	0.8284	-0.0091	0.0101
Crippa/Tukey	0.5233	-0.0032	0.0036	1.4436	-0.0278	0.0296
BLM types	0.4448	-0.0017	0.0024	1.0050	-0.0064	0.0087
Low-rank factors	0.4073	-0.0022	0.0028	0.9074	-0.0115	0.0128
GKLP	0.4135	-0.0030	0.0034	1.0074	-0.0175	0.0186

BLM.

- Grouped-minus-full gaps are zero under the BLM DGP because its schedule already has two worker types and three firm classes.
- Under continuous low rank, the gaps are -1.233 (Q_F), -2.479 (H_F), -0.545 (ρ_H), and -0.200 (C^w_{assign}).
- Therefore accurate estimation of the grouped object need not recover a continuous schedule.

4.5 Rank selection

Figure 4 separates BIC rank selection from numerical status. Correct dimension does not imply stable factors or schedule completion.

Table 9: BIC selection counts and separate project diagnostic-pass rate

DGP	True	Select 0	Select 1	Select 2	Pass
AKM	0	100	0	0	100%
Crippa/Tukey	1	0	100	0	39%
BLM types	1	0	100	0	85%
Low-rank factors	2	0	0	100	93%
GKLP	1	0	100	0	100%

- BIC selects the true rank in all 500 populations: 0 for AKM; 1 for Crippa, BLM, and GKLP; 2 for continuous low rank.
- The larger, more mobile design resolves earlier small-design rank errors.
- Pass rates still differ. Correct rank and reproducible completion are different events.

4.6 Numerical warnings and squared-error tails

Figure 5 compares unconditional all-returned RMSE with RMSE among diagnostic passes. RMSE squares errors, so a few extreme completions can dominate it. Pass-only is a sensitivity check, not an alternative estimator.

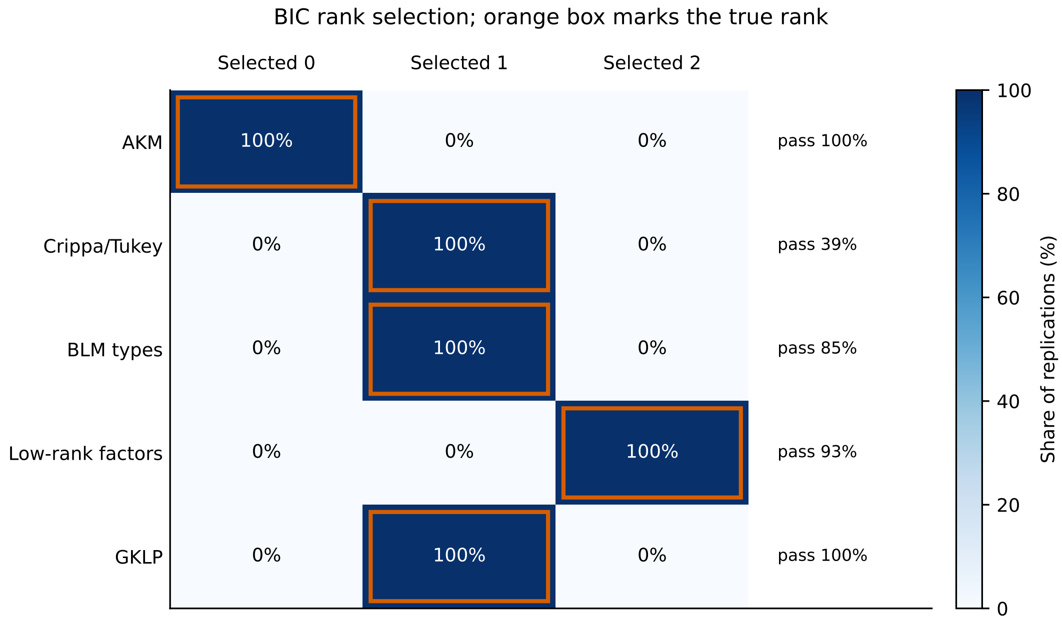

Figure 4: BIC rank selection. The orange outline marks the DGP's true rank; the label at right is the distinct numerical pass rate. The figure is included to separate model-selection performance from optimizer stability.

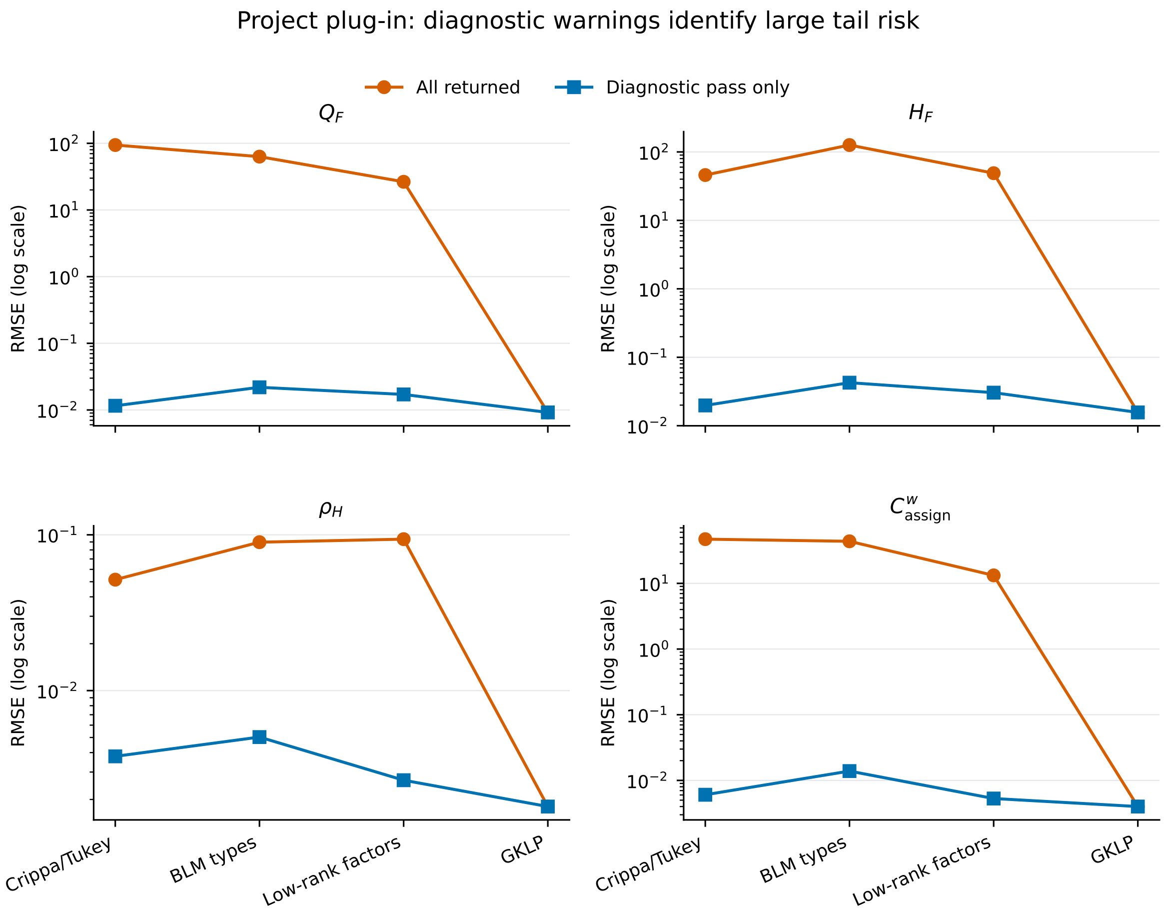

Figure 5: Project plug-in RMSE among all returned values and conditional on a numerical pass. The logarithmic scale shows that warning fits, rather than ordinary sampling noise, generate the large unconditional errors.

Tail concentration for Q_F:

- Median absolute error is 0.0114 (Crippa), 0.0105 (BLM), and 0.0159 (continuous low rank).
- Maximum absolute error is 480, 498, and 230, respectively.
- The five largest replications produce 91.1%, 100%, and 96.9% of total squared error. Typical accuracy therefore coexists with poor unconditional risk.

Numerical evidence:

- **Continuous low rank:** median worst local condition number is 42 for 93 passes and 84,006 for 7 warnings; median maximum factor norm is 4.32 versus 184.9.
- **BLM:** corresponding condition numbers are 157 and 6,340.
- **Crippa:** pass and warning medians are 98 and 117, but the warning 90th percentile is 2.53 million; the warning 90th-percentile maximum factor norm is 154. Some warnings are mild; a smaller subset drives the RMSE tail.

Mechanism:

1. At rank r, each worker ALS update regresses on $[1, v_j']$ for visited firms.
2. Visited-firm vectors may be nearly collinear. The local Gram matrix is then nearly singular, and its pseudoinverse amplifies small match-mean noise.
3. Large worker or firm factors propagate to unobserved pairs during completion. More periods can add information; centering changes only normalization and cannot add independent local factor variation.

5 Interpretation

1. **Keep DGP and estimator separate.** KSS succeeds under AKM; BLM succeeds under the discrete BLM DGP. Other rows measure the cost of their restrictions.
2. **Keep native and project targets separate.** KSS estimates an additive projection; BS20 estimates C_{BS}, not C^{w}_{assign}; BLM compresses a continuous schedule into six cells. Common-target RMSE is meaningful only after these mappings and gaps are stated.
3. **Rank selection is not the remaining problem.** BIC selects the true rank in all populations. Completion can still be locally ill-conditioned at the correct rank.
4. **The current plug-in is a baseline, not the final LOO procedure.** The completion step needs a declared safeguard, such as regularized local regressions, conditioning-aware support, or direct estimation of target functionals. Section 3.5 separates baseline calibration from untested variants. No current result shows that a proposed regularization improves efficacy.

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